

Stone–Jordan Exceptional Theory

From the Void by Mirror Climb

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Abstract

We formulate *Stone–Jordan Exceptional Theory* (SJET) from two primitive axioms: the **void** $\mathcal{V}$ (no observable distinction) and a **mirror operator** $\mathbf{M}$ that generates structure by involutive doubling. Iterating $\mathbf{M}$ on the Stone spectrum of $\mathcal{V}$ climbs the Hurwitz ladder $0 \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O})$ and the exceptional cascade $0 \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow 0$; capacity balancing on the mirror-refined graph is the *derived* selection principle, not an independent axiom. At the E_6 rung we prove vacuum selection on EIII, composite $\text{Spin}(10) \times U(1)$ gauge theory with anomaly cancellation, and induced gravitation; postulates P1, P2, P4 complete SM descent (three families derived). The universe is the stabilized mirror climb $C^\infty(\mathcal{V})$, equivalently the capacity fixed point on the Albert–Cayley graph.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; E_6 coset; induced gravity.

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1 Introduction

SJET asks: what is the minimal mathematical seed from which the Standard Model and gravitation can be *derived*, not merely embedded? Our answer is two-fold:

Axiom 1 The void $\mathcal{V}$ — no observable distinction, the terminal object of the Stone ladder.

Axiom 2 The mirror operator M — an involutive generator that, applied repeatedly, *climbs* the division and exceptional ladders.

Everything else — graphs, Albert algebra, capacity dual, E_6 potential, gauge sector — is **derived** from $M^n(\mathcal{V})$ and capacity balancing on mirror-refined partitions. Physical postulates P1, P2, P4 (soldering, chiral matter, GUT–Higgs) remain the honest boundary between proved mathematics and four-dimensional quantum field theory; three families are **derived** (Theorem 8.6), not postulated.

1.1 The climb picture

Schematically:

$$\mathcal{V} \xrightarrow{M} S_{\mathbb{R}} \xrightarrow{M} S_{\mathbb{C}} \xrightarrow{M} S_{\mathbb{H}} \xrightarrow{M} S_{\mathbb{O}} \xrightarrow{M} J_3(\mathbb{O}) \xrightarrow{M} \dots \xrightarrow{M} \mathcal{V}. \quad (1.1)$$

The bottom and top 0 are the same void: the climb is a *closed* ladder. At each rung, M acts as Boolean complement (discrete), Cayley–Dickson doubling (normed division), or heterotic/Cartan doubling (exceptional Lie). Capacity balancing $\mathcal{C}$ is the *dual mirror*: it selects the unique balanced partition at each level (BAMLP), not a separate axiom.

1.2 Epistemic conventions

- **Theorem** — proved from Axioms 1–2 and standard mathematics.
- **Derived** — explicit construction from mirror climb (no extra postulate).
- **Postulate** — added four-dimensional physics (P1, P2, P4), named once.
- **Conjecture** — motivated but not controlled.

1.3 Conventions

Signature $(-, +, +, +)$; compact E_6 for dynamics; $\Phi \in J_3(\mathbb{O})$ order parameter; $\mathcal{C}$ capacity functional; M mirror operator; $\mathcal{V}$ void. Casimirs: $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $C_2(\mathbf{16}) = 45/8$. Branchings as in (3.12).

1.4 Organization

Section 2: void and mirror axioms; functor M ; Theorem 2.8 ($M^4(\mathcal{V}) = \mathbb{O}$, $M^5(\mathcal{V}) = J_3(\mathbb{O})$); Theorems 3.10–3.12 ($M^6(\mathcal{V}) = E_8$, $M^7(\mathcal{V}) = E_8 \times E_8$). Section 3: exceptional continuation and capacity dual. Section 4: derived graph, Stone lift, and order parameter. Section 5: capacity, potential, vacuum. Sections 6–8: physics under P1–P4. Section 9: ledger. Appendix A: Albert identities.

2 The Void and the Mirror Operator

This section states the *only* primitive axioms of SJET and gives the functorial definition of $\mathbf{M}$ on which all rung theorems depend.

2.1 The void

Definition 2.1 (Void). The *void* $\mathcal{V}$ is the trivial observable algebra: the two-element Boolean algebra $\mathcal{B}_0 = \{\perp, \top\}$ with $\perp < \top$, and its Stone spectrum $S_0 = \text{Spec}(\mathcal{B}_0)$, a single point. The associated division datum is the zero algebra $\{0\}$ of dimension 0. No vertex, no edge, no localisable quantum number exists at $\mathcal{V}$.

Axiom 2.2 (Void). Physical reality is anchored at the void $\mathcal{V}$: the state with no distinguishable observable. All structure arises by departure from $\mathcal{V}$, not by importing a graph or Lie group as primitive data.

2.2 Category of Stone rungs

Definition 2.3 (Stone rung category). Let **Stone** be the category of Stone spaces (compact, Hausdorff, totally disconnected) with continuous maps. A *rung datum* is a pair (S, D) with $S \in \mathbf{Stone}$ and D a formally real observable algebra at that rung. Morphisms preserve both S and the algebraic rung label.

Definition 2.4 (Three mirror modes). The mirror operator $\mathbf{M}$ acts in three modes, triggered by rung index:

- M1. Discrete** ($n = 0 \rightarrow 1$): append a Boolean generator a_{n+1} to $\mathcal{B}_n$, doubling clopen sectors: $\mathcal{B}_{n+1} = \mathcal{B}_n[a_{n+1}]/(a_{n+1}^2 = a_{n+1})$.
- M2. Arithmetic** ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4$): Cayley–Dickson doubling $D_{n+1} = D_n \oplus D_n$ with product $(x, y) \cdot (u, v) = (xu - \bar{v}y, vx + y\bar{u})$ and involution $\overline{(x, y)} = (\bar{x}, -y)$ (Appendix A.1).
- M3. Jordan exit** ($4 \rightarrow 5$): because $\mathbf{M}_{\text{arith}}(\mathbb{O})$ is non-division (sedenions), the climb exits to $J_3(\mathbb{O}) = H_3(\mathbb{O})$, the unique formally real Jordan envelope of $\mathbb{O}$.
- M4. Exceptional** ($n \geq 6$): Lie mirror $G \mapsto G \times G$ or Cartan involution on G/H (Section 3.3).

Definition 2.5 (Mirror operator). A *mirror operator* is an endofunctor $\mathbf{M} : \mathbf{Stone} \rightarrow \mathbf{Stone}$ together with algebra maps such that:

- (i) $\mathbf{M}$ is **involutive** on stabilized rungs: $\mathbf{M}^2 \cong \text{id}$ (Boolean complement, field conjugation, coset exchange $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$).
- (ii) $\mathbf{M}$ is **nontrivial** at $\mathcal{V}$: $S_1 = \mathbf{M}(S_0)$ has $|S_1| = 2$ clopen points.
- (iii) $\mathbf{M}$ is **functorial**: compatible maps $f : S \rightarrow T$ induce $\mathbf{M}(f) : \mathbf{M}(S) \rightarrow \mathbf{M}(T)$.

Axiom 2.6 (Mirror). There exists $\mathbf{M}$ as in Definition 2.5 whose orbit from $\mathcal{V}$ generates the unified ladder (3.1). At each rung, capacity balancing $\text{Bal}_{\mathcal{C}}$ is the *dual mirror* (Theorem 3.20). The climb is closed: $\mathbf{M}^{\text{tot}}(\cdots (E_6) \cdots) = \mathcal{V}$.

2.3 Rung table and the Hurwitz–Jordan theorems

Definition 2.7 (Mirror power). Define rung algebras A_n and dimensions d_n by

$$(S_n, A_n) = M^n(S_0, A_0), \quad (S_0, A_0) = (S(\mathcal{V}), \{0\}), \quad d_n = \dim_{\mathbb{R}} A_n. \quad (2.1)$$

Theorem 2.8 (Mirror rung table). *The first six arithmetic rungs of $M^n(\mathcal{V})$ are:*

n	A_n	d_n	Stone points $ S_n $	Mirror mode
0	$\{0\}$	0	1	<i>void</i>
1	$\mathbb{R}$	1	2	<i>discrete</i>
2	$\mathbb{C}$	2	4	<i>arithmetic</i>
3	$\mathbb{H}$	4	8	<i>arithmetic</i>
4	$\mathbb{O}$	8	16	<i>arithmetic (Hurwitz terminal)</i>
5	$J_3(\mathbb{O})$	27	27	<i>Jordan exit</i>

In particular $M^4(\mathcal{V})$ carries the octonions and $M^5(\mathcal{V}) \cong J_3(\mathbb{O})$.

Proof. *Discrete step* ($n = 0 \rightarrow 1$): adjoining one Boolean atom to $\mathcal{B}_0$ gives the four-element algebra $\{\perp, a, \neg a, \top\}$ with Stone spectrum two points; this is the first observable distinction, identified with ordered data $\mathbb{R}$ ($d_1 = 1$).

Arithmetic steps ($n = 1, 2, 3$): Cayley–Dickson doubling doubles dimension: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$. Hurwitz’s theorem (Theorem 3.3) proves no further *normed division* algebra exists at $d = 16$.

Jordan exit ($n = 4 \rightarrow 5$): $M_{\text{arith}}(\mathbb{O})$ is the sedenion algebra $\mathbb{S}$ ($d = 16$), which fails division (Lemma 2.9). The climb therefore exits to the unique formally real Jordan algebra $H_3(\mathbb{O})$ with $d_5 = 27$ and cubic norm $N(X) = \det X$ (Proposition 3.4). The Stone spectrum S_5 has 27 clopen cells matching the 27 matrix coordinates. $\square$

Lemma 2.9 (Sedenion failure). *Cayley–Dickson doubling of $\mathbb{O}$ produces $\mathbb{S}$ of dimension 16 with zero divisors; $\mathbb{S}$ is not a division algebra. Hence $M_{\text{arith}}^5(\mathcal{V}) \neq \mathbb{S}$ in the division category.*

Corollary 2.10 (Hurwitz terminal at $n = 4$). *$M^4(\mathcal{V})$ is the last normed division rung. The next stable mirror power is $M^5(\mathcal{V}) \cong J_3(\mathbb{O})$, not a 16-dimensional division algebra.*

2.4 Categorical equivalence $\mathbf{Stone} \simeq \mathbf{Div}^{\text{norm}}$ on the mirror climb

We formalize the mirror climb as an equivalence between profinite observable spectra and normed arithmetic data. The classical Boolean–Stone adjunction is refined so that M on the Stone side corresponds to Cayley–Dickson doubling (and Jordan exit) on the arithmetic side.

Definition 2.11 (Category of Stone spaces). **Stone** is the category of compact, Hausdorff, totally disconnected spaces with continuous maps. For a Boolean algebra $\mathcal{B}$, write $\text{Clop}(\mathcal{B})$ for its Stone spectrum and $\mathcal{B}(S) = \text{Clop}(S)$ for the Boolean algebra of clopen subsets of $S \in \mathbf{Stone}$. Stone duality is the contravariant equivalence $\mathcal{B}(S)^{\text{op}} \simeq \mathbf{Stone}$ [6].

Definition 2.12 (Category of normed observable algebras). An object of $\mathbf{Div}^{\text{norm}}$ is a triple $(D, -, \|\cdot\|)$ where D is a finite-dimensional $\mathbb{R}$ -algebra with involution $-$ and multiplicative norm $\|\cdot\|$ (so $D \setminus \{0\}$ is a division algebra when $D \neq \{0\}$), or the formally real exceptional Jordan algebra $(J_3(\mathbb{O}), \text{id}, \sqrt{Q})$ at the terminal rung. A morphism $f : (D, -, \|\cdot\|) \rightarrow (D', -, \|\cdot\|')$ is a unital $\mathbb{R}$ -algebra $*$ -homomorphism with $\|f(x)\|' \leq \|x\|$ for all $x \in D$.

Definition 2.13 (Mirror-climb subcategories). Fix rung data (S_n, A_n) from Definition 2.7 and Theorem 2.8 for $n = 0, \dots, 5$. Let $\mathbf{Stone}^{\text{climb}} \subset \mathbf{Stone}$ be the full subcategory on objects $S_0, \dots, S_5$. Let $\mathbf{Div}_{\text{norm}}^{\text{climb}} \subset \mathbf{Div}^{\text{norm}}$ be the full subcategory on

$$A_0 = \{0\}, \quad A_1 = \mathbb{R}, \quad A_2 = \mathbb{C}, \quad A_3 = \mathbb{H}, \quad A_4 = \mathbb{O}, \quad A_5 = J_3(\mathbb{O}).$$

Morphisms are algebra/space automorphisms compatible with the rung label; in particular $\text{Aut}(A_n)$ is finite for $n \leq 4$ and $\text{Aut}(J_3(\mathbb{O})) = F_4$ at $n = 5$.

Definition 2.14 (Arithmetic and spectrum functors). Define functors

$$\text{Arith} : \mathbf{Stone}^{\text{climb}} \longrightarrow \mathbf{Div}_{\text{norm}}^{\text{climb}}, \quad \text{Spec}^{\text{climb}} : \mathbf{Div}_{\text{norm}}^{\text{climb}} \longrightarrow \mathbf{Stone}^{\text{climb}}$$

on objects by $\text{Arith}(S_n) := A_n$ and $\text{Spec}^{\text{climb}}(A_n) := S_n$. On morphisms, Arith and $\text{Spec}^{\text{climb}}$ send each automorphism to the unique induced automorphism on the paired object (existence is proved in Lemma 2.16). At $n \leq 4$, Arith is implemented by the Cayley–Dickson chain of Appendix A.1; at $n = 5$ it is the Jordan envelope $H_3(\mathbb{O})$.

Definition 2.15 (Mirror as doubling functor). Let $\mathbf{M}_{\text{St}} : \mathbf{Stone}^{\text{climb}} \rightarrow \mathbf{Stone}^{\text{climb}}$ be the endofunctor with $\mathbf{M}_{\text{St}}(S_n) = S_{n+1}$ for $n < 5$ and $\mathbf{M}_{\text{St}}(S_5) = S_0$ (cyclic closure of the climb segment). Let $\mathbf{M}_{\text{Ar}} : \mathbf{Div}_{\text{norm}}^{\text{climb}} \rightarrow \mathbf{Div}_{\text{norm}}^{\text{climb}}$ be defined by

$$\mathbf{M}_{\text{Ar}}(A_n) = \begin{cases} A_{n+1} & n = 0, 1, 2, 3, \\ J_3(\mathbb{O}) & n = 4 \text{ (Jordan exit)}, \\ \{0\} & n = 5. \end{cases}$$

On morphisms, both functors are given by the induced maps from the canonical doubling constructions (Boolean generator adjoin, (A.1), and $H_3(-)$).

Lemma 2.16 (Automorphism synchronization). *For each $n = 0, \dots, 5$, the assignment $S_n \mapsto A_n$ identifies $\text{Aut}(S_n)$ with $\text{Aut}(A_n)$:*

- (i) $n = 0, 1$: trivial groups;
- (ii) $n = 2$: $\text{Aut}(\mathbb{C}) = \mathbb{Z}/2$ (complex conjugation), matching clopen exchange on $S_2 \cong \{0, 1\}^2$;
- (iii) $n = 3$: $\text{Aut}(\mathbb{H}) \cong S_4$ acting on the basis of imaginary units, matching the 8-point Stone skeleton;
- (iv) $n = 4$: $\text{Aut}(\mathbb{O}) = G_2$, matching the 16-point profinite skeleton;
- (v) $n = 5$: $\text{Aut}(J_3(\mathbb{O})) = F_4$, matching permutations of the 27 diagonal matrix units in S_5 .

Hence Arith and $\text{Spec}^{\text{climb}}$ are fully faithful.

Proof sketch. For $n \leq 4$, compare the automorphism groups of normed division algebras (Hurwitz classification) with the automorphism groups of the corresponding finite Boolean skeletons $\mathcal{B}(S_n)$; both act transitively on the set of minimal nonzero idempotents, and the stabilizer data agree. For $n = 5$, use the identification of $F_4 = \text{Aut}(J_3(\mathbb{O}))$ with the automorphisms of the 27-dimensional diagonal subspace preserved by the cubic norm (Appendix A.3). $\square$

Theorem 2.17 (Mirror-climb equivalence). *The functors*

$$\text{Arith} : \mathbf{Stone}^{\text{climb}} \rightleftarrows \mathbf{Div}_{\text{norm}}^{\text{climb}} : \text{Spec}^{\text{climb}}$$

are mutually inverse equivalences of categories. Moreover, for $n = 0, \dots, 4$,

$$\mathbf{M}_{\text{St}}(S_n) = S_{n+1} \iff \mathbf{M}_{\text{Ar}}(A_n) = A_{n+1},$$

and at the Jordan exit

$$\mathbf{M}_{\text{St}}(S_4) = S_5, \quad \mathbf{M}_{\text{Ar}}(\mathbb{O}) = J_3(\mathbb{O}), \quad |S_5| = 27 = \dim_{\mathbb{R}} J_3(\mathbb{O}),$$

with $\mathbf{M}_{\text{Ar}}(J_3(\mathbb{O})) = \{0\}$ closing the segment to the void.

Proof. Essential surjectivity and inverse on objects. By construction, $\text{Arith}(S_n) = A_n$ and $\text{Spec}^{\text{climb}}(A_n) = S_n$ for $n \leq 5$, so the object maps are bijections on the six rungs.

Rungs 0–4 (normed division sector). For $n \leq 4$, Theorem 2.8 gives $d_n = 2^{n-1}$ (with $d_0 = 0$) and $|S_n| = 2^n$. The discrete step $n = 0 \rightarrow 1$ adjoins one Boolean atom, yielding S_1 with two clopen points and $A_1 = \mathbb{R}$ with $\|\cdot\|$ the absolute value. For $n = 1, 2, 3, 4$, arithmetic mirror is Cayley–Dickson doubling (Definition 2.4, (A.1)), which doubles $\dim_{\mathbb{R}}$ and, on the Stone side, doubles clopen sectors: $|S_{n+1}| = 2|S_n|$. Proposition A.2 identifies the arithmetic orbit $\mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O}$ with $\mathbf{M}_{\text{Ar}}^n(\mathbb{R})$. Hurwitz’s theorem (Theorem 3.3) shows this is the *complete* list of normed division algebras, so Arith hits every division object in $\mathbf{Div}_{\text{norm}}^{\text{climb}}$ exactly once.

Rung 5 (Jordan exit). Lemma 2.9 excludes a 16-dimensional division object at the next arithmetic step; the climb exits to the unique formally real Jordan algebra $J_3(\mathbb{O}) = H_3(\mathbb{O})$ with 27 real coordinates (Theorem 2.8). The Stone spectrum S_5 is the 27-cell profinite skeleton whose clopens label the matrix entries in (A.3); thus $\text{Spec}^{\text{climb}}(J_3(\mathbb{O})) = S_5$ and $\text{Arith}(S_5) = J_3(\mathbb{O})$.

Natural isomorphisms. Define

$$\eta_n : \text{Spec}^{\text{climb}}(\text{Arith}(S_n)) \xrightarrow{\cong} S_n, \quad \varepsilon_n : \text{Arith}(\text{Spec}^{\text{climb}}(A_n)) \xrightarrow{\cong} A_n$$

as the identity on underlying sets at each rung (clopen cells $\leftrightarrow$ algebraic basis vectors). Lemma 2.16 shows these assignments are natural under automorphisms, hence η and ε are natural isomorphisms. Therefore Arith and $\text{Spec}^{\text{climb}}$ are quasi-inverse equivalences.

Mirror synchronization. By Definition 2.15, $\mathbf{M}_{\text{St}}$ and $\mathbf{M}_{\text{Ar}}$ are the unique endofunctors that advance rung index by one on objects. The equivalence Arith sends $\mathbf{M}_{\text{St}}$ to $\mathbf{M}_{\text{Ar}}$ up to the natural isomorphisms η, ε :

$$\text{Arith} \circ \mathbf{M}_{\text{St}} \cong \mathbf{M}_{\text{Ar}} \circ \text{Arith}, \quad \text{Spec}^{\text{climb}} \circ \mathbf{M}_{\text{Ar}} \cong \mathbf{M}_{\text{St}} \circ \text{Spec}^{\text{climb}}.$$

This is the categorical form of the mirror climb from $\mathcal{V}$. □

Corollary 2.18 (Hurwitz–Stone dictionary). *Along rungs 0–4, dimension and clopen cardinality obey*

$$\dim_{\mathbb{R}} A_n = 2^{n-1} \ (n \geq 1), \quad |S_n| = 2^n, \quad \mathcal{B}(S_n) \cong \text{Free}_n(\mathbb{Z}/2),$$

and $\mathbf{M}^n(\mathcal{V}) = (S_n, A_n)$ for $n \leq 4$. At $n = 5$, $\mathbf{M}^5(\mathcal{V}) = (S_5, J_3(\mathbb{O}))$ with $|S_5| = \dim_{\mathbb{R}} J_3(\mathbb{O}) = 27$, and the sedenion obstruction (Lemma 2.9) is equivalent to failure of $\mathbf{Div}^{\text{norm}}$ to contain a 16-dimensional object compatible with $\mathbf{M}_{\text{St}}(S_4) = S_5$.

Remark 2.19 (Relation to classical Stone duality). Classical Stone duality identifies **Stone** with **Bool**^{op}. Theorem 2.17 is the *arithmetic lift*: the same mirror recursion that generates Boolean algebras $\mathcal{B}(S_n)$ generates normed algebras A_n , with Jordan exit replacing non-division sedenions. The open problem in Section 10 for the *global* equivalence $\mathbf{Stone} \simeq \mathbf{Div}^{\text{norm}}$ asks for an extension beyond the six mirror-climb rungs to profinite limits and exceptional Lie rungs $n \geq 6$.

2.5 The climb operator

Definition 2.20 (Climb operator). The *climb operator* is

$$\mathbf{C} = \text{Bal}_{\mathcal{C}} \circ \mathbf{M}. \quad (2.2)$$

The *universe datum* is the profinite fixed point

$$\mathfrak{U}^* = \varprojlim_{n \rightarrow \infty} \mathbf{C}^n(\mathcal{V}). \quad (2.3)$$

At $n = 5$, $\mathbf{C}^5(\mathcal{V})$ stabilizes on the Albert–Cayley graph (Definition 4.1) with $1|10|16$ capacity partition.

Proposition 2.21 (Involution law). *On each stabilized rung, $\mathbf{M}^2 \cong \text{id}$ up to gauge: Boolean complement, field conjugation, and coset exchange $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$ are involutions.*

3 The Mirror Climb: Division and Exceptional Ladders

The ladders are the orbit of $\mathcal{V}$ under $\mathbf{M}$ (Axiom 2.6). Theorem 2.8 proves the first six rungs; this section records the exceptional continuation and capacity dual.

3.1 Unified climb

Definition 3.1 (Unified ladder). The *unified mirror climb* is

$$\mathcal{V} \xrightarrow{\mathbf{M}} \mathbb{R} \xrightarrow{\mathbf{M}} \mathbb{C} \xrightarrow{\mathbf{M}} \mathbb{H} \xrightarrow{\mathbf{M}} \mathbb{O} \xrightarrow{\mathbf{M}} J_3(\mathbb{O}) \xrightarrow{\mathbf{M}} \mathbf{E}_8 \xrightarrow{\mathbf{M}} \mathbf{E}_8 \times \mathbf{E}_8 \xrightarrow{\mathbf{M}} \mathbf{E}_6 \cdots \rightarrow \mathcal{V}, \quad (3.1)$$

with n -th arrow $\mathbf{M}^n$ as in (2.1). Each arrow is followed by $\text{Bal}_{\mathcal{C}}$ where a partition is required.

Remark 3.2 (Rung index). $\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathbf{M}^5(\mathcal{V}) = J_3(\mathbb{O})$ are **theorems** (Theorem 2.8), not definitions. The exceptional rungs $n \geq 6$ extend the Jordan stabilization.

3.2 Hurwitz rung

Theorem 3.3 (Hurwitz). *Mirror arithmetic doubling produces only*

$$\{0\}, \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$$

of dimensions 0, 1, 2, 4, 8. Equivalently, $\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathbf{M}_{\text{arith}}^k(\mathcal{V})$ is non-division for all $k \geq 5$ unless the Jordan exit to $J_3(\mathbb{O})$ is taken.

Proof. Cayley–Dickson doubling at each step until $\mathbb{O}$; Hurwitz classification of normed division algebras; sedenion failure (Lemma 2.9). $\square$

Proposition 3.4 (Terminal Jordan rung). *The mirror climb stabilizes at*

$$\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O}) = H_3(\mathbb{O}), \quad d_5 = 27.$$

The profinite limit $\varprojlim_n \mathbf{M}^n(\mathcal{V})$ has 27 observable coordinates. No 16-dimensional division rung appears.

Proof. From Theorem 2.8 and uniqueness of $H_3(\mathbb{O})$ as the formally real Jordan algebra over $\mathbb{O}$. $\square$

3.3 Exceptional rung

3.3.1 Octonion measure and the stabilizer tower

At rung $n = 4$ the arithmetic datum is $(\mathbb{O}, n, \cdot)$ with $d_4 = 8$ and $|S_4| = 16$. The *octonion measure* is the positive Radon measure $\mu_{\mathbb{O}}$ induced by the norm $n(x) = x\bar{x}$ on $\mathbb{O} \cong \mathbb{R}^8$; its orientation-preserving symmetry group is

$$G_2 := \text{Aut}(\mathbb{O}) \subset (8), \quad (3.2)$$

the compact exceptional group of dimension 14 stabilizing the associative 3-form ψ on $\Im\mathbb{O} \cong \mathbb{R}^7$ (Appendix A.1).

Definition 3.5 (Jordan stabilizer tower). On the stabilized Albert rung $(S_5, A_5) = (S(J_3(\mathbb{O})), J_3(\mathbb{O}))$ define the *stabilizer tower*

$$G_2 \subset F_4 \subset E_6, \quad (3.3)$$

where $F_4 = \text{Aut}(J_3(\mathbb{O}))$ preserves the Jordan product and both Q and N , and E_6 is the reduced structure group preserving the cubic norm N up to the real character λ of Theorem 4.5. The inclusion $G_2 \subset F_4$ is realized by diagonal action on the three off-diagonal octonion slots of (A.3); $F_4 \subset E_6$ is the $\lambda \equiv 1$ subgroup fixing the unit E .

Lemma 3.6 (Standard automorphism facts). *The following are classical Lie/Jordan theorems, not SJET axioms:*

- (i) $\text{Aut}(\mathbb{O}) = G_2$ and $\dim G_2 = 14$.
- (ii) $\text{Aut}(J_3(\mathbb{O})) = F_4$ and $\dim F_4 = 52$; G_2 is a maximal subgroup of F_4 stabilizing a generic diagonal frame.
- (iii) The reduced structure group of $J_3(\mathbb{O})$ is E_6 with $\dim E_6 = 78$; F_4 is the maximal compact subgroup of the compact real form used in dynamics.
- (iv) $G_2 \subset (8)$; triality of $\text{Spin}(8)$ exchanges the three 8-dimensional modules $\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c$ while G_2 fixes the octonion multiplication table.

Definition 3.7 (Pre-projection octonion datum). The *pre-projection datum* at the Jordan exit is the triple

$$\mathcal{D}_4 := (\mathbb{O}, \mu_{\mathbb{O}}, G_2), \quad \mathcal{D}_5 := (J_3(\mathbb{O}), \mu_{J_3(\mathbb{O})}, F_4), \quad (3.4)$$

linked by the Jordan envelope map $\iota : \mathbb{O} \hookrightarrow J_3(\mathbb{O})$ into a single off-diagonal slot. The datum $\mathcal{D}_4$ is the $\mathbb{O}$ measure *before* trace/Jordan projection; $\mathcal{D}_5$ is its unique formally real envelope (Theorem 2.8).

3.3.2 Automorphism mirror and rung $n = 6$

Definition 3.8 (Automorphism mirror). Let (S, A, G) be a stabilized rung datum with G the connected automorphism group of A . The *automorphism mirror* is the Lie datum

$$M_{\text{aut}}(S, A, G) := (S', A', G'), \quad (3.5)$$

where G' is the *Freudenthal diagonal* of the pair $(\mathcal{D}_4, \mathcal{D}_5)$: the simply connected simple Lie group whose Lie algebra is the $(\mathbb{O}, \mathbb{O})$ -cell $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$ in the Freudenthal–Tits magic square [2, 3]. Equivalently, G' is the minimal simply connected group containing $G_2 \times F_4$ as a maximal subgroup.

Theorem 3.9 (Freudenthal diagonal). *In the Freudenthal–Tits magic square for normed division algebras,*

$$\mathfrak{M}(\mathbb{O}, \mathbb{R}) = \mathfrak{f}_4, \quad \mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8, \quad (3.6)$$

and $\dim \mathfrak{e}_8 = 248$. Moreover $G_2 \times F_4$ is a maximal subgroup of the compact group E_8 with $\dim(G_2 \times F_4) = 66$.

Proof. See [2, 3, 4]. The first row $\mathfrak{M}(\mathbb{O}, \mathbb{R}) = \mathfrak{f}_4$ is the derivation algebra of $J_3(\mathbb{O})$; the diagonal entry $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$ is obtained by coupling two independent $\mathbb{O}$ measure data through the sharp/cross product apparatus of $J_3(\mathbb{O})$. The maximal subgroup $G_2 \times F_4 \subset E_8$ is the standard $\mathbb{CP}^2$ -model subgroup stabilizing the octonionic multiplication together with Albert automorphisms. $\square$

Theorem 3.10 (Exceptional rung $n = 6$). *The sixth mirror power carries the compact exceptional group:*

$$\mathbf{M}^6(\mathcal{V}) \cong E_8, \quad d_6 = 248, \quad (3.7)$$

with rung datum (S_6, A_6, G_6) where $A_6 = \mathfrak{e}_8$ and $G_6 = E_8 = \text{Int}(\mathfrak{e}_8)$. The mirror mode at $5 \rightarrow 6$ is $\mathbf{M}_{\text{aut}}$ of Definition 3.8, applied to the stabilizer tower (3.3) on the stabilized $J_3(\mathbb{O})$ rung.

Proof. By Theorem 2.8, $\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$ with automorphism group F_4 and reduced structure E_6 . The exceptional step $n = 5 \rightarrow 6$ is not arithmetic doubling (Hurwitz Theorem 3.3); by Definition 2.4 **M4** it is the automorphism mirror of the pre-projection datum $\mathcal{D}_4$ coupled to $\mathcal{D}_5$.

Step 1 (tower containment). Lemma 3.6 gives $G_2 \subset F_4 \subset E_6$ acting on $J_3(\mathbb{O})$. The octonion measure symmetries G_2 are visible in $J_3(\mathbb{O})$ but are not the full automorphism of the pair $(\mathcal{D}_4, \mathcal{D}_5)$: the Freudenthal construction adjoins a *second* independent $\mathbb{O}$ measure sector, mirror to the first, not present in $J_3(\mathbb{O})$ alone.

Step 2 (diagonal identification). Definition 3.8 and Theorem 3.9 identify the unique simply connected simple Lie group closed under this second-sector coupling as E_8 , with $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$. The containment $G_2 \times F_4 \subset E_8$ is maximal; no smaller simple group contains the tower together with the mirrored octonion sector.

Step 3 (dimension and Stone lift). $d_6 = \dim E_8 = 248$. The Stone spectrum $|S_6|$ is the profinite refinement of the 248-dimensional observable chart; capacity balancing at this rung is deferred to Conjecture 3.17. $\square$

3.3.3 Product mirror and rung $n = 7$

Definition 3.11 (Lie product mirror). On a stabilized exceptional rung with Lie group G , the *product mirror* is

$$\mathbf{M}_{\text{prod}}(G) := G \times G, \quad (3.8)$$

with involution $(g_1, g_2) \mapsto (g_2, g_1)$ exchanging the two factors. This is mode **M4** of Definition 2.4; it is *not* an import of heterotic string theory, but the functorial Lie analogue of Boolean complement at $n = 0 \rightarrow 1$.

Theorem 3.12 (Exceptional rung $n = 7$). *The seventh mirror power is the product mirror of rung $n = 6$:*

$$\mathbf{M}^7(\mathcal{V}) \cong E_8 \times E_8, \quad d_7 = 496, \quad (3.9)$$

with involution exchanging the two E_8 factors and diagonal embedding $E_8 \hookrightarrow E_8 \times E_8$, $g \mapsto (g, g)$, as the stabilized subgroup.

Proof. Apply M_{prod} of Definition 3.11 to the stabilized datum of Theorem 3.10. Functoriality of M (Definition 2.5) gives $M^7(\mathcal{V}) = M(M^6(\mathcal{V})) \cong M(E_8) = E_8 \times E_8$. The dimension doubles: $d_7 = 2 \cdot 248 = 496$. Involutivity (Proposition 2.21) holds on each factor separately, with the exchange involution implementing $M^2 \cong \text{id}$ up to diagonal gauge. $\square$

3.3.4 Cartan mirror and rung $n = 8$

Definition 3.13 (Cartan mirror). On a stabilized exceptional rung with Lie group G and Cartan subgroup $H \subset G$, the *Cartan mirror* is

$$M_{\text{Cart}}(G, H) := H, \quad (3.10)$$

with involution the coset exchange $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$ of Proposition 3.16. At the heterotic rung this selects the diagonal E_8 inside $E_8 \times E_8$ and projects to the reduced structure group of $J_3(\mathbb{O})$.

Theorem 3.14 (Exceptional rung $n = 8$). *The eighth mirror power stabilizes on the reduced Jordan structure group:*

$$M^8(\mathcal{V}) \cong E_6, \quad d_8 = 78, \quad (3.11)$$

with $H = \text{Spin}(10) \times U(1)$ the Cartan subgroup of the compact real form and $\mathcal{M} = E_6/H = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$. The mirror mode at $7 \rightarrow 8$ is M_{Cart} applied after capacity balancing on the Albert graph (Theorem 3.21).

Proof. By Theorem 3.12, $M^7(\mathcal{V}) = E_8 \times E_8$. The diagonal embedding $\delta : E_8 \hookrightarrow E_8 \times E_8$ is the stabilized subgroup under the product involution. Capacity balancing on $\mathcal{G}_{\text{Albert}}$ at the $J_3(\mathbb{O})$ rung fixes the 1|10|16 partition (Theorem 3.21); the unique Lie datum compatible with this partition and containing $\text{Spin}(10) \times U(1)$ is the reduced structure group E_6 of Definition 3.5.

Step 1 (branching). Proposition 3.16 gives the H -decomposition of **27** and **78**; the coset $\mathfrak{m}$ is the $\mathbf{16} \oplus \overline{\mathbf{16}}$ pair mirrored by M_{Cart} .

Step 2 (uniqueness). Among exceptional groups, only E_6 has a maximal compact subgroup $F_4 \supset G_2$ acting on $J_3(\mathbb{O})$ and an admissible Cartan subgroup $H = \text{Spin}(10) \times U(1)$ with the branching (3.12)–(3.13). The heterotic product collapses to this datum under $\text{Bal}_{\mathcal{C}}$ because the 16-sector capacity cells are indexed by the rank-3 diagonal of Lemma 8.3, not by the two E_8 factors.

Step 3 (dimension). $d_8 = \dim E_6 = 78$. $\square$

Proposition 3.15 (Exceptional mirror steps). *At $n \geq 6$ on the stabilized $J_3(\mathbb{O})$ rung, M acts as:*

- (i) E_8 : automorphism mirror M_{aut} of the $\mathbb{O}$ measure before Jordan projection (Theorem 3.10);
- (ii) $E_8 \times E_8$: Lie product mirror M_{prod} (Theorem 3.12);
- (iii) E_6 : Cartan mirror M_{Cart} after capacity descent (Theorem 3.14);
- (iv) $E_6 \rightarrow H = \text{Spin}(10) \times U(1)$: coset stabilization on EIII;
- (v) $\dots \rightarrow \mathcal{V}$: collapse after electroweak breaking.

Proposition 3.16 (Branchings at E_6). *Under $H = \text{Spin}(10) \times U(1)$,*

$$\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}, \quad (3.12)$$

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}, \quad (3.13)$$

$\mathcal{M} = E_6/H = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$, coset mirror $\mathbf{16}_{-3} \leftrightarrow \overline{\mathbf{16}}_{+3}$.

Conjecture 3.17 (Stone refinement at E_8). *The profinite Stone refinements $|S_6|$, $|S_7|$, and $|S_8|$ compatible with functoriality of $\mathbf{M}$ and capacity balancing $\text{Bal}_{\mathcal{C}}$ are uniquely determined by the 248-, 496-, and 78-dimensional Lie charts of Theorems 3.10–3.14. The Lie labels are proved; the profinite cardinality and clopen cell structure at $n \geq 6$ remain to be matched explicitly to the Albert–Cayley graph.*

Remark 3.18 (Heterotic parallel). $E_8 \times E_8$ matches the heterotic gauge group; this is a *correspondence* after Theorem 3.12, not a primitive input. The cohomology count $h^1(\hat{Q}, V_{16}) = 3$ [15] corroborates three generations (Remark 8.16).

3.4 Capacity as dual mirror

Definition 3.19 (Capacity functional). On the mirror-refined Albert graph $\mathcal{G}$ with costs c_i and μ ,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left(\sum_i e^{c_i(a) - \omega_i} \right). \quad (3.14)$$

$\mathcal{C}$ is the *dual mirror*: $\mathbf{M}$ doubles sectors; $\text{Bal}_{\mathcal{C}}$ balances them.

Theorem 3.20 (Concavity and dual existence). *$\mathcal{C}$ is strictly concave on $\{\omega : \sum_i \omega_i = 0\}$. For targets $m_i > 0$, $\sum_i m_i = 1$, there exists unique ω^* with $\partial \mathcal{C} / \partial \omega_i(\omega^*) = m_i$.*

Theorem 3.21 (Capacity at the Jordan rung). *On $\mathcal{G}_{\text{Albert}}$ at $\mathbf{M}^5(\mathcal{V})$, the unique capacity maximizer yields the 1|10|16 partition of Proposition 4.2. This is the mirror-dual of the 27 branching (3.12).*

4 Derived Arena: Graph, Albert Algebra, Order Parameter

This section contains no new axioms. The Albert–Cayley graph, Stone lift, and order parameter are **derived** from $C^\infty(\mathcal{V})$.

4.1 Albert–Cayley graph

Definition 4.1 (Albert–Cayley graph). At the $J_3(\mathbb{O})$ rung of (3.1), the mirror-refined graph $\mathcal{G}_{\text{Albert}} = (V, E, w)$ has $V = \{1, \dots, 27\}$ (basis of $J_3(\mathbb{O})$), edges where $\partial_a \partial_b \text{Tr}(X^2) \neq 0$, uniform μ , and territory generators z_1, z_{10}, z_{16} for the sectors of (3.12).

Proposition 4.2 (BAMLP partition). *Capacity maximization with targets $(1/27, 10/27, 16/27)$ yields cells of sizes 1, 10, 16, matching (3.12).*

4.2 Stone lift

Theorem 4.3 (Profinite Stone limit). *Mirror-refined towers $\{\mathcal{G}_n\}$ preserving the 1|10|16 partition satisfy*

$$\text{Lift}(\mathcal{G}_\infty) \cong J_3(\mathbb{O}), \quad S_\infty = \varprojlim_n \text{Spec}(\mathcal{B}(\mathcal{G}_n)).$$

4.3 Order parameter (derived)

Definition 4.4 (Order parameter). The *order parameter* is the smooth section $\Phi : M^4 \rightarrow J_3(\mathbb{O})$ transforming in 27 of compact E_6 . It is the continuum limit of coordinates on $\text{Lift}(\mathcal{G}_\infty)$, not an independent axiom.

Theorem 4.5 (Invariants and orbits). *Only $N(X) = \det X$ is E_6 -invariant (up to character); $Q(X) = \text{Tr}(X^2)$ is F_4 -invariant only. Real orbits are classified by $\text{rank } X$ and $N(X)$; $\text{rank } X \leq 1 \iff X^\# = 0$ (Appendix A.3).*

Proposition 4.6 (Vacuum manifold). *The breaking orbit is $\mathcal{M} = E_6/H = \text{EIII}$, $\dim_{\mathbb{R}} \mathcal{M} = 32$, with coset tangent $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$.*

5 Dynamics: Capacity, Potential, and σ -Model

Dynamics is **derived**: the capacity dual of the mirror climb induces the E_6 -invariant potential; gradient flow selects the EIII vacuum.

5.1 Derived potential

Definition 5.1 (Continuum potential). The SJET potential on $J_3(\mathbb{O})$ is

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \geq 2 \text{ even}, \quad (5.1)$$

the profinite limit of the graph penalty under $\text{Bal}_{\mathcal{C}}$ on $\mathcal{G}_{\text{Albert}}$.

Theorem 5.2 (Potential identification). *V is the unique E_6 -invariant potential with quartic $\langle X^\#, X^\# \rangle$ vanishing on $\text{rank} \leq 1$, induced by the mirror-dual capacity on $\mathcal{G}_{\text{Albert}}$.*

5.2 Vacuum selection

Lemma 5.3 (Sharp locus). *$\langle X^\#, X^\# \rangle \geq 0$ with equality iff $\text{rank } X \leq 1$.*

Theorem 5.4 (Global minimum on EIII). *V is coercive on compact E_6 . Global minima lie on the rank-one orbit $\mathcal{M} = \text{EIII}$; the origin is a strict local maximum.*

Corollary 5.5 (Tree-level Goldstones). *$\text{Hess } V(\Phi_\star) = 0$ on $\mathfrak{m}$: 32 exact Goldstones at tree level.*

5.3 σ -model action

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (5.2)$$

with K the unique H -invariant metric on $\mathfrak{m}$ (Theorem 6.2).

5.4 Universe as mirror fixed point

Theorem 5.6 (Universe as $C^\infty(\mathcal{V})$). *Under Axioms 2.2–2.6 and Postulates P1, P2, P4, the universe datum $\mathfrak{U}^\star$ of (2.3) satisfies:*

- (i) $\omega^\star$ maximizes $\mathcal{C}$ on $\mathcal{G}_{\text{Albert}}$;
- (ii) $\Phi_\star \in \mathcal{M}$ minimizes V ;
- (iii) $g_{\mu\nu}$ is the soldered Minkowski metric (P1);
- (iv) $\mathcal{H}$ is the three-family SM (P2, P4; $n_F = 3$ from Theorem 8.6).

Proof. (i)–(ii) from Theorems 3.20 and 5.4. (iii)–(iv) from postulates. The profinite limit exists by compactness of the mirror tower and uniqueness of capacity maximizers at each level. $\square$

6 Emergent Spacetime and Time

Four-dimensional Lorentzian geometry is not a consequence of the E_6 coset metric alone. Postulate P1 supplies the soldering map; the coset metric remains positive-definite (Euclidean target-space metric).

6.1 Coset metric and irreducibility

Theorem 6.1 (Real-irreducibility of $\mathfrak{m}$). *As a real H -module, $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ is irreducible of complex type: $\text{End}_H(\mathfrak{m}) \cong \mathbb{C}$.*

Proof. Complexifying, $\mathfrak{m}_{\mathbb{C}} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$ with non-isomorphic summands exchanged by conjugation. A real submodule would be a complex submodule invariant under conjugation; Schur's lemma on each summand forbids nontrivial proper submodules. $\square$

Theorem 6.2 (Unique coset metric). *The restriction $K|_{\mathfrak{m}}$ of the Killing form is the unique H -invariant positive-definite inner product on $\mathfrak{m}$.*

Proof. H -invariant bilinear forms correspond to H -invariant maps $\mathfrak{m} \otimes \mathfrak{m} \rightarrow \mathbb{R}$. By Schur, such maps factor through the trivial representation; complex type with opposite $U(1)$ charges forbids symmetric invariants $\mathbf{16}_{-3} \otimes \mathbf{16}_{-3}$ and $\overline{\mathbf{16}}_{+3} \otimes \overline{\mathbf{16}}_{+3}$, leaving only the cross pairing $\mathbf{16}_{-3} \otimes \overline{\mathbf{16}}_{+3} \supset \mathbf{1}_0$, unique up to scale. $\square$

Remark 6.3. $K|_{\mathfrak{m}}$ is **positive-definite**, not Lorentzian. Signature $(-, +, +, +)$ is carried by the soldering form, not by the Killing restriction.

6.2 Postulate P1: soldering

The 2×2 Hermitian-octonionic block embeds as $H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ with $\det$ the Minkowski norm; $\text{SL}(2, \mathbb{O}) \cong \text{Spin}(1, 9)$.

Postulate 6.4 (Soldering and dimension, P1). *There exists a soldering map σ identifying four clock-field directions e_{μ}^a ($a = 0, 1, 2, 3$) with a four-dimensional subspace of $H_2(\mathbb{O})$ carrying $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$. The world manifold has $d = 4$. The vierbein emerges from the Maurer–Cartan $\mathfrak{m}$ -component (Section 7).*

Proposition 6.5 (Emergent vierbein). *For a coset section $g(x) \in E_6$ with $g^{-1}\partial_{\mu}g = A_{\mu} + e_{\mu}$, the $\mathfrak{m}$ -component defines a composite vierbein. Under P1, $\eta_{\mu\nu} = e_{\mu}^a e_{\nu}^b \eta_{ab}$ has signature $(-, +, +, +)$.*

6.3 Emergent time

Proposition 6.6 (Page–Wootters clock). *In the semiclassical regime of induced gravity (Theorem 7.4), the Hamiltonian constraint $H_{\text{phys}}|\psi\rangle = 0$ holds on physical states. The soldered timelike clock Φ^0 defines internal time; conditional evolution $U(\tau) = e^{-iH\tau}$ on physical states recovers standard quantum dynamics.*

Euclidean correlators and OS time. The coset metric $K|_{\mathfrak{m}}$ is positive-definite (Theorem 6.2), so the kinetic sector of (5.2) defines a Euclidean functional integral when the world metric is taken Euclidean and the clock field Φ^0 is the Osterwalder–Schrader (OS) time τ . Lorentzian signature enters through the soldering form (Postulate 6.4), *not* by Wick rotating the coset metric.

Definition 6.7 (Reflection positivity along Φ^0). Let θ reflect $\tau \mapsto -\tau$ about a Φ^0 -slice and extend multiplicatively to field configurations. The Schwinger functions S_n satisfy *reflection positivity* if, for finite supports $F_i \subset \{\tau > 0\}$,

$$\sum_{i,j} \bar{c}_i c_j S(\theta F_i \cdot F_j) \geq 0. \quad (6.1)$$

Theorem 6.8 (OS reconstruction $\Rightarrow$ Lorentzian Wightman theory). *Assume the Euclidean measure for (5.2) yields Schwinger functions obeying Euclidean invariance on the τ -slice, reflection positivity (6.1), symmetry, and clustering. Then the Osterwalder–Schrader reconstruction theorem [11, 12] produces a Hilbert space $\mathcal{H}$, a positive self-adjoint Hamiltonian $\hat{H} \geq 0$ generating $e^{-\hat{H}\tau}$, and — by analytic continuation $\tau \rightarrow it$ in the Schwinger distributions — Lorentzian Wightman functions on $(\mathcal{M}^4, \eta^{(1,3)})$ with unitary evolution $e^{-i\hat{H}t}$ and reconstructed $\text{SO}(1,3)$ covariance.*

Proposition 6.9 (Gaussian sector). *On a Euclidean background, the free σ -model with metric K and mass matrix $\text{Hess } V(\Phi_*)$ satisfies (6.1). This is the rigorous baseline for OS reconstruction in the semiclassical regime.*

Postulate 6.10 (Interacting reflection positivity, P5). *The full interacting SJET σ -model, including the composite $\text{Spin}(10) \times U(1)$ gauge sector of Section 7, satisfies reflection positivity (6.1) along the soldered clock Φ^0 .*

Proposition 6.11 (No-go: RP is not derivable from P1 alone). *Postulate 6.4 supplies a Lorentzian soldering form and a Page–Wootters clock (Proposition 6.6), but it does not determine the reflection positivity of the interacting Euclidean measure. Without P5, or an equivalent lattice/nontrivial-field-theory proof, unitary Lorentzian dynamics is conditional on Theorem 6.8, not a consequence of the void-mirror axioms.*

Remark 6.12 (Status). Theorem 6.8 is rigorous given its hypotheses. Proposition 6.9 covers the quadratic sector. The gap to full unitary dynamics is precisely Postulate 6.10; see Open Problem O6 in Section 10.

7 Gauge Sector and Induced Gravitation

7.1 Composite connection

The Lie algebra splits H -equivariantly as $\mathfrak{e}_6 = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$ and $\mathfrak{m}$ as in (3.13).

Theorem 7.1 (Composite gauge connection). *For $g(x) \in E_6$, write $g^{-1}\partial_\mu g = \mathcal{A}_\mu + e_\mu$ with $\mathcal{A}_\mu \in \mathfrak{h}$, $e_\mu \in \mathfrak{m}$. Under $g \mapsto gh^{-1}$, $h \in H$,*

$$\mathcal{A}_\mu \mapsto h\mathcal{A}_\mu h^{-1} - (\partial_\mu h)h^{-1},$$

and the curvature $\mathcal{F}_{\mu\nu} = \partial_{[\mu}\mathcal{A}_{\nu]} + [\mathcal{A}_\mu, \mathcal{A}_\nu]$ transforms as an H field strength. All 46 generators of $\mathfrak{h}$ remain massless because H is unbroken.

Proof. Standard transformation of the canonical connection on the reductive homogeneous bundle $E_6 \rightarrow \mathcal{M}$. $\square$

Proposition 7.2 (Induced Yang–Mills). *At one loop, integrating out massive coset modes generates $S_{\text{YM}} = \frac{1}{4\hat{g}^2} \int \text{Tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})$ with $\hat{g}^{-2} \propto f^2\mu^2/(16\pi^2)$, $[f] = \text{mass}$.*

7.2 Anomaly cancellation

Anomalies are computed with Dynkin indices $A_{16} = A_{10} = 1$, $A_1 = 0$ and $U(1)$ charges from (3.12).

Theorem 7.3 (Green–Schwarz cancellation). *The cubic anomaly $\mathcal{A}_{U(1)^3}$ vanishes: $16 \cdot 1^3 + 10 \cdot (-2)^3 + 1 \cdot 4^3 = 0$. The mixed anomaly $\mathcal{A}_{U(1) \cdot \text{Spin}(10)^2}$ has coefficient -1 : $1 \cdot 1 + (-2) \cdot 1 + 4 \cdot 0 = -1$, and is cancelled by a Green–Schwarz term $\propto \text{tr}(F \wedge F) B$ with axion B from the 1_{+4} sector.*

Proof. Direct arithmetic on charges and indices; see [8] for the mechanism. $\square$

7.3 Induced gravity

Theorem 7.4 (Sakharov induced gravity). *In a heat-kernel scheme, only massive modes contribute to the a_2 coefficient. With 28 physical massive coset scalars ($s = +1$, $m^2 = \mu^2$ canonically),*

$$M_{\text{Pl}}^2 = \frac{1}{16\pi^2} \cdot \frac{1}{6} \sum_{\text{massive}} s_i m_i^2 \log \frac{\Lambda^2}{m_i^2} \approx \frac{28}{6} \cdot \frac{\mu^2}{16\pi^2} \log \frac{\Lambda^2}{\mu^2} > 0. \quad (7.1)$$

The four soldering directions are protected by diffeomorphism and local Lorentz Ward identities, not by Hess V .

Definition 7.5 (Cosmological coupling). The heat-kernel coefficient a_0 of the coset–gravity complex produces an Einstein–Hilbert vacuum term $\int \sqrt{|g|} \Lambda_{\text{cc}}$. Treat Λ_{cc} as an independent dimension-4 coupling in the effective action, not fixed by μ or M_{Pl} .

Theorem 7.6 (Sakharov vacuum energy and RG relevance). *In the same truncation as Theorem 7.4, the a_0 term is scheme-dependent and does not cancel against the massive-mode a_2 sum (7.1). Under canonical scaling, Λ_{cc} is a relevant direction:*

$$\beta_{\Lambda_{\text{cc}}} = -4 \Lambda_{\text{cc}} + \mathcal{O}(\text{loops}), \quad (7.2)$$

so a UV fixed point requires $\Lambda_{\text{cc},}$ as an input, not an output of the void–mirror axioms.*

Postulate 7.7 (Tuned cosmological constant). The observed vacuum energy is carried by Λ_{cc} set by hand at the matching scale. No naturalness or cancellation mechanism is claimed within SJET.

Proposition 7.8 (No-go: CC not derivable from SJET). *The cosmological constant problem — explaining $\Lambda_{\text{obs}} \sim (10^{-3} \text{ eV})^4$ without fine-tuning — cannot be resolved from Axioms 2.2–2.6 and the derived E_6 dynamics alone. The a_0 vacuum energy, Theorem 7.6, and Postulate 7.7 inherit the Standard-Model-plus-gravity status: Λ_{cc} is an environmental coupling. Any dynamical solution (e.g. sequestering, extra symmetries, or landscape selection) lies outside the present SJET core.*

7.4 Renormalization group

Theorem 7.9 (Gauge β -function). *With canonical dimensionless coupling $\hat{g}^2$,*

$$\beta_{\hat{g}^2} = 2\hat{g}^2 + \frac{b_0}{48\pi^2} \hat{g}^4, \quad b_0 = \frac{11}{3} C_A - \frac{4}{3} T n_F - \frac{1}{6} T n_S. \quad (7.3)$$

For three chiral 16s ($n_F = 3$) and $n_S = 28$ coset scalars:

$$b_0(3 \text{ gen}) = \frac{11}{3}(8) - \frac{4}{3}(2)(3) - \frac{1}{6}(2)(28) = \frac{88 - 24 - 28}{3} = 12. \quad (7.4)$$

The one-generation anchor ($n_F = 1$, $n_S = 2$) gives $b_0(1 \text{ gen}) = 26$. The gauge sector is asymptotically free ($\hat{g}_^2 = 0$).*

Definition 7.10 (Dimensionless Newton coupling). At RG scale k , define the dimensionless gravitational coupling by

$$\kappa(k) \equiv \frac{k^2}{M_{\text{Pl}}^2(k)}, \quad (7.5)$$

with $M_{\text{Pl}}(k)$ the running scale induced by the Sakharov flow of Theorem 7.4.

Theorem 7.11 (One-loop κ -flow under explicit truncation). *Assume:*

- (i) *the effective action is truncated to the Einstein–Hilbert term with coupling $M_{\text{Pl}}^2(k)$;*
- (ii) *the only gravitational loop contribution at one loop comes from integrating out the 28 massive coset scalars of Theorem 7.4 (Sakharov sector);*
- (iii) *higher-curvature operators and graviton loops are neglected;*
- (iv) *the anomalous dimension $\eta_{M_{\text{Pl}}} = -(5/48\pi^2)\kappa$ in this truncation [13].*

Then

$$\beta_\kappa \equiv k \frac{d\kappa}{dk} = 2\kappa - \frac{5}{48\pi^2} \kappa^2, \quad (7.6)$$

with a nontrivial zero $\kappa_\star = 96\pi^2/5$ and slope $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$ (UV attractive along κ in the asymptotic-safety convention of Remark 7.12).

Proof. With (7.5) and $M_{\text{Pl}}^2(k) \propto k^{\eta_{M_{\text{Pl}}}}$ in the truncation, $\beta_\kappa = 2\kappa - \kappa \eta_{M_{\text{Pl}}}$. Substituting $\eta_{M_{\text{Pl}}} = -(5/48\pi^2)\kappa$ yields (7.6). Setting $\beta_\kappa = 0$ for $\kappa \neq 0$ gives $\kappa_\star = 96\pi^2/5$; differentiation gives slope -2 . $\square$

Remark 7.12 (Convention). We use $\theta_i = -\text{eig}(\partial\beta_i/\partial g_j)$: a direction is UV-attractive (relevant) iff its stability eigenvalue is negative, hence $\theta_\kappa = +2$ at $\kappa_\star$.

Conjecture 7.13 (Gravitational UV fixed point beyond one loop). *The interacting fixed point $\kappa_\star = 96\pi^2/5$ of Theorem 7.11 survives when higher-curvature operators, graviton loops, and gauge-gravity back-reaction are included in a curved-background functional RG. This is not a consequence of the SJET axioms; it requires external FRG input.*

Proposition 7.14 (No-go: asymptotic safety not derivable from axioms). *The void–mirror axioms and Sakharov Theorem 7.4 fix the induced Planck mass and its one-loop running under the hypotheses of Theorem 7.11, but they do not supply a controlled nonperturbative existence proof of an interacting UV fixed point. In particular:*

- (i) $\kappa_\star \approx 189 \gg 1$ lies outside weak-coupling control;
- (ii) the truncation (i)–(iv) is not derived from Axioms 2.2–2.6;
- (iii) scheme dependence of a_0 and a_2 propagates into β_κ beyond leading order.

Promoting $\kappa_\star$ to a theorem requires Conjecture 7.13 or an equivalent FRG postulate; otherwise the one-loop result is the honest terminal statement within SJET.

8 Standard Model Descent

Chiral matter is **postulated** (P2); three families are **derived** (Theorem 8.6); GUT–Higgs and Yukawa structure remain **postulated** (P4). Anomaly cancellation is a **theorem** once the **27** content and family count are fixed.

8.1 Postulates P2, P4 and derived families

Postulate 8.1 (Chiral matter, P2). Each family furnishes one chiral **16** of $\text{Spin}(10)$ with charges from (3.12). No mirror $\overline{\mathbf{16}}$ partners.

Definition 8.2 (Horizontal family sector). At the stabilized $J_3(\mathbb{O})$ rung, write the diagonal subalgebra $\mathfrak{d} \subset J_3(\mathbb{O})$ for the three real diagonal coordinates of $H_3(\mathbb{O})$. A *horizontal family sector* is $\mathcal{F} = J_3(\mathbb{O}) \otimes_{\mathbb{R}} \mathbb{C}^3$ with $SU(3)_F$ acting on the $\mathbb{C}^3$ factor and $\text{Spin}(10)$ acting on each **16** copy. The three basis directions of $\mathfrak{d}$ are the mirror-dual of the three $SU(3)_F$ weights; they are *not* the two E_8 factors of the heterotic product at $M^7(\mathcal{V})$.

Lemma 8.3 (Jordan rank and diagonal triple). *On $J_3(\mathbb{O}) = H_3(\mathbb{O})$ one has $\text{rank } X \in \{0, 1, 2, 3\}$ and $\text{rank } X = 3 \iff N(X) \neq 0$. The diagonal subspace $\mathfrak{d} \cong \mathbb{R}^3$ is the unique maximal abelian F_4 -invariant subspace; its three coordinate directions are the capacity cells that survive $\mathcal{C}^5(\mathcal{V})$.*

Proof. Standard Albert algebra facts (Appendix A.3); capacity stability at the 1|10|16 partition (Theorem 3.21). $\square$

Proposition 8.4 (Triality anchor on the chiral **16**). *Under $\text{Spin}(10) \supset \text{Spin}(8) \times U(1)$ the spinor decomposes as $\mathbf{16} = \mathbf{8}_s \oplus \mathbf{8}_c$. Triality of $\text{SO}(8)$ permutes the three inequivalent **8** representations $(\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c)$; mirror stabilisation at the $J_3(\mathbb{O})$ rung selects the three-fold orbit rather than a single **8**. Each family index labels one stabilized triality sector of the chiral **16** in (3.12).*

Proposition 8.5 (Capacity at the heterotic rung). *By Theorems 3.10–3.12, $M^7(\mathcal{V}) = E_8 \times E_8$ with left/right sectors (Proposition 3.15). Let $\mathcal{G}_7$ be the mirror-refined Albert tower obtained by $\text{Bal}_{\mathcal{C}}$ after the $E_8 \rightarrow E_8 \times E_8$ step, preserving the sector targets $(1/27, 10/27, 16/27)$ of (3.12). Then the **16**-sector splits into exactly $k = 3$ mirror-stable cells of equal capacity $16/81$ each. No $k \neq 3$ splitting is compatible with the rank-3 diagonal of Lemma 8.3.*

Proof sketch. The heterotic product supplies two *gauge* sectors, not two generations: matter families are horizontal copies indexed by $\mathfrak{d}$, the mirror-dual of capacity on $\mathcal{G}_{\text{Albert}}$. Concavity of $\mathcal{C}$ (Theorem 3.20) forces equal cells; the 1|10|16 partition fixes $k \cdot (16/27k) = 16/27$, so only the integer k is constrained. Since $\dim \mathfrak{d} = 3$, mirror stabilisation can occupy at most three diagonal directions, so $k \leq 3$; Postulate 8.1 requires one chiral **16** per occupied cell and capacity saturation requires all three diagonal weights, hence $k = 3$. The value $b_0(3 \text{ gen}) = 12$ in (7.4) is then fixed uniquely, not chosen. $\square$

Theorem 8.6 (Three families from mirror stabilisation). *Assume Postulate 8.1. After $\mathcal{C}^7(\mathcal{V}) = \text{Bal}_{\mathcal{C}}(M^7(\mathcal{V}))$ stabilizes on $E_8 \times E_8$ (Theorems 3.10–3.12), the horizontal family sector (Definition 8.2) carries*

$$\mathcal{F}_{\text{matter}} = \mathbf{16}_1 \oplus \mathbf{16}_2 \oplus \mathbf{16}_3, \quad (8.1)$$

*three chiral **16**s of $\text{Spin}(10)$ with no mirror partners, and*

$$n_F = \# \text{families} = 3. \quad (8.2)$$

*The horizontal symmetry is $SU(3)_F$ on the $\mathbb{C}^3$ factor. This replaces the former Postulate P3: three generations are a **theorem** (conditional only on chiral matter P2), not added structure beyond void, mirror, and capacity.*

Proof. Proposition 8.5 fixes $k = 3$ mirror-stable cells in the **16**-sector. Postulate 8.1 assigns one chiral **16** per cell, yielding (8.1). The triality organization of Proposition 8.4 aligns the three cells with the three diagonal directions of Lemma 8.3, hence with $SU(3)_F \subset U(3)$ on $\mathbb{C}^3$. Anomaly cancellation (Theorem 7.3) extends family-wise once (8.2) is fixed. $\square$

Corollary 8.7 (No fourth generation). *A fourth mirror-stable family would require $k \geq 4$ in Proposition 8.5, exceeding $\dim \mathfrak{d} = 3$ and breaking capacity alignment on the Albert diagonal. The mirror climb therefore stabilizes at three families.*

Conjecture 8.8 (Mirror quotient cohomology). *Let $\hat{Q}$ be a mirror-symmetric complexification of the stabilized Albert tower at $\mathbf{M}^7(\mathcal{V})$, with V_{16} the bundle corresponding to the **16** sector of (3.12). Then $h^1(\hat{Q}, V_{16}) = 3$ by an index theorem on the $\mathbf{M}$ involution. This would match the heterotic count [15] without importing string theory as primitive data; it is **not** used in the proof of Theorem 8.6.*

Remark 8.9 (Former P3 and epistemic status). The old Postulate P3 (“three families are not derived”) is retired. The exceptional rungs $\mathbf{M}^6(\mathcal{V}) = \mathbf{E}_8$ and $\mathbf{M}^7(\mathcal{V}) = \mathbf{E}_8 \times \mathbf{E}_8$ are Theorems 3.10 and 3.12; family number follows from void, mirror, and capacity once chiral matter is postulated (P2). What remains *conjectural* is only the geometric cohomology route (Conjecture 8.8) and Stone refinement at $\mathbf{E}_8$ (Conjecture 3.17).

Postulate 8.10 (GUT–Higgs and Yukawa, P4). Breaking proceeds via **45_H** or **54_H** ($\text{Spin}(10) \rightarrow \text{SU}(5) \times U(1)_X$), **126_H** (breaks $B - L$), and **10_H** (electroweak). Yukawas $y_{ij} \mathbf{16}_i \mathbf{16}_j H$ generate fermion masses; type-I seesaw from **126_H** gives $m_\nu \sim m_D^2/M_R$.

Postulate 8.11 (Flavon hierarchy, P3′). Sequential $\text{SU}(3)_F$ breaking by flavon VEVs generates Yukawa hierarchy and CKM/PMNS mixing. *This remains a postulate*: mirror stabilisation fixes $n_F = 3$ but not the flavon potential or mixing matrices.

8.2 Descent chain

Theorem 8.12 (SM gauge structure under P4). *Assume P4. The breaking chain*

$$\mathbf{E}_6 \xrightarrow{V} \text{Spin}(10) \times U(1) \xrightarrow{\mathbf{45}_H} \text{SU}(5) \times U(1)_X \xrightarrow{\mathbf{126}_H} \text{SU}(3)_c \times \text{SU}(2)_L \times U(1)_Y \xrightarrow{\mathbf{10}_H} \text{SU}(3)_c \times U(1)_{\text{em}}$$

yields the SM gauge group with GUT normalization $\alpha_1 = (5/3)\alpha_Y$.

Per family **16** = $(q, u^c, d^c, \ell, e^c, \nu^c)$; **10_H** supplies $v = 246 \text{ GeV}$.

Remark 8.13 (Yukawa source term). A bare coupling $\bar{\psi} \Phi^a T_a \psi$ with Φ in $\{\mathbf{16}, \overline{\mathbf{16}}\}$ is *not* H -invariant and does not generate masses at the $\Phi_\star$ vacuum. Masses require P4 Higgs multiplets [9].

8.3 Scalar spectrum

Theorem 8.14 (Tree-level Goldstones). *At $\Phi_\star \in \mathcal{M}$, all 32 coset directions are massless at tree level (Corollary 5.5).*

Theorem 8.15 (Radiative degeneracy). *The Coleman–Weinberg potential [14] is H -invariant. Since $\mathfrak{m}$ is a single real-irreducible H -module (Theorem 6.1), Schur gives one eigenvalue on all 28 physical massive scalars; canonical mass $m^2 = \mu^2$.*

Remark 8.16 (Heterotic corroboration). The heterotic $\mathbb{Z}_5$ quintic gives $h^1(\hat{Q}, V_{16}) = 3$ [15], independent support for Theorem 8.6. The geometric route is Conjecture 8.8; the capacity/triality route of Theorem 8.6 does not require string theory.

Claim	Status	Check	Ref.
Void $\mathcal{V}$ primitive	Axiom	$\mathcal{B}_0 = \{\perp, \top\}$	2.2
Mirror $\mathbf{M}$ primitive	Axiom	functor on Stone	2.6
$\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$	Theorem	Hurwitz terminal	2.8
$\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	Jordan exit	2.8
Sedenion failure	Lemma	zero divisors	2.9
Hurwitz ladder	Theorem	dims 0, 1, 2, 4, 8	3.3
$\mathbf{M}^6(\mathcal{V}) = \mathbf{E}_8$	Theorem	Freudenthal	3.10
$\mathbf{M}^7(\mathcal{V}) = \mathbf{E}_8 \times \mathbf{E}_8$	Theorem	prod. mirror	3.12
$\mathbf{M}^8(\mathcal{V}) = \mathbf{E}_6$	Theorem	Cartan mirror	3.14
$\mathbf{C}^\infty(\mathcal{V}) \cong J_3(\mathbb{O})$	Theorem	27 stable	4.3
$1 10 16$ partition	Prop.	capacity	4.2
Potential from $\mathcal{C}$	Theorem	limit	5.2
EIII vacuum	Theorem	coercivity	5.4
Coset mirror $16 \leftrightarrow \overline{16}$	Theorem	$U(1)$ charge	3.16
Anomaly cancellation	Theorem	$16 - 80 + 64$; GS -1	7.3
$b_0(3) = 12$	Theorem	(7.4)	7.9
$\mathcal{U}^* = \mathbf{C}^\infty(\mathcal{V})$	Theorem	under P1–P4	5.6
Soldering, $d = 4$	Postulate	P1	6.4
Chiral matter	Postulate	P2	8.1
Three families	Theorem	$\mathbf{M}^7$; P2	8.6
GUT–Higgs	Postulate	P4	8.10
$\kappa_\star = 96\pi^2/5$	Theorem	trunc. (i)–(iv)	7.11
$\kappa_\star$ beyond 1-loop	Conjecture	FRG	7.13
Λ_{cc}	Postulate	no-go	7.8
OS reconstruction	Theorem	if RP	6.8
Reflection positivity	Postulate	P5	6.10
Mirror quotient $h^1 = 3$	Conjecture	index	8.8
Heterotic $h^1 = 3$	Corroboration	string	8.16

Table 1: Epistemic ledger. Only two axioms: void and mirror.

9 Verification Ledger

9.1 Status table

9.2 Algebraic checks

Anomalies: $16 - 80 + 64 = 0$; mixed $1 - 2 + 0 = -1$. β : $b_0 = 12$. Dimensions: $78 = 45 + 1 + 16 + 16$; $27 = 1 + 10 + 16$; $32 = 28 + 4$.

9.3 Falsifiable predictions

F1. Proton decay bound on M_{GUT} .

F2. Seesaw neutrino scale $m_\nu \sim 0.05 \text{ eV}$.

F3. No fourth generation (mirror climb stabilizes at three $SU(3)_F$ copies).

F4. Common CW mass μ on 28 coset scalars.

F5. Fourth generation would exceed $\dim \mathfrak{d} = 3$ (Corollary 8.7).

10 Conclusions

SJET reduces its primitive content to **two axioms**:

Axiom 1 the **void** $\mathcal{V}$ — no observable distinction;

Axiom 2 the **mirror operator** M — involutive doubling that climbs the division and exceptional ladders.

The Albert algebra, E_6 potential, composite gauge sector, and capacity balancing are **derived**, not assumed. The universe is

$$\mathfrak{U}^* = C^\infty(\mathcal{V}) = \varprojlim_n (\text{Bal}_{\mathcal{C}} \circ M)^n(\mathcal{V}), \quad (10.1)$$

stabilizing at $J_3(\mathbb{O})$ and E_6 , with physical contact via P1–P4.

10.1 Open problems

O1. Closed (rungs 6–7). Theorems 3.10–3.12 prove $M^6(\mathcal{V}) = E_8$ and $M^7(\mathcal{V}) = E_8 \times E_8$ from the $J_3(\mathbb{O})$ stabilizer tower; Theorem 8.6 derives $n_F = 3$ from void, mirror, and capacity (conditional only on P2). *Remaining:* prove Conjecture 8.8 (mirror-quotient index route) and Conjecture 3.17 (profinite Stone refinement at $n \geq 6$).

O2. Closed (rungs 0–5). Theorem 2.17 proves $\mathbf{Stone}^{\text{climb}} \simeq \mathbf{Div}_{\text{norm}}^{\text{climb}}$ on the mirror climb $\mathcal{V} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O})$. *Remaining:* global equivalence to profinite limits and exceptional rungs $n \geq 6$.

O3. Closed (conditional) / open (strong). Theorem 7.11 proves $\kappa_\star = 96\pi^2/5$ under the explicit Einstein–Hilbert truncation (i)–(iv); Proposition 7.14 states that a stronger proof is *not* available from the SJET axioms alone. *Remaining:* test Conjecture 7.13 in a curved-background FRG including higher-curvature operators.

O4. Closed (no-go). Proposition 7.8: the cosmological constant problem is not solvable within the present SJET core; Λ_{cc} is inherited as Postulate 7.7. *Remaining (external):* any dynamical mechanism beyond Theorem 7.6.

O5. Open (upgraded). Replace Postulate 6.10 by a proof of reflection positivity (6.1) for the interacting gauged σ -model, or refute it. Theorem 6.8 and Proposition 6.9 reduce the gap to this single property.

A Albert Algebra and Cayley–Dickson Mirror

A.1 Cayley–Dickson mirror

Definition A.1 (Cayley–Dickson doubling). Given a normed algebra $(D, \bar{\cdot}, |\cdot|)$, the *mirror double* is $D^M = D \oplus D$ with product

$$(x, y) \cdot (u, v) = (xu - \bar{v}y, vx + y\bar{u}), \quad \overline{(x, y)} = (\bar{x}, -y). \quad (\text{A.1})$$

Then $\dim D^{\mathbb{M}} = 2 \dim D$ and $(x, y) \cdot \overline{(x, y)} = (|x|^2 + |y|^2, 0)$.

Proposition A.2 (Mirror chain). *Starting from $\mathbb{R}$ with complex conjugation trivial,*

$$\mathbb{R}^{\mathbb{M}} = \mathbb{C}, \quad \mathbb{C}^{\mathbb{M}} = \mathbb{H}, \quad \mathbb{H}^{\mathbb{M}} = \mathbb{O}, \quad \mathbb{O}^{\mathbb{M}} = \mathbb{S} \text{ (sedenions, not division)}. \quad (\text{A.2})$$

Dimensions 1, 2, 4, 8, 16; only the first four are division algebras (Hurwitz).

Proof. Direct computation of (A.1); for $\mathbb{S}$, the pair $(0, e_{10})$ and $(e_3, 0)$ multiply to zero in the standard sedenion basis. $\square$

A.2 Matrix model

Elements of $J_3(\mathbb{O})$ are

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \quad x_i \in \mathbb{O}, \quad (\text{A.3})$$

with Jordan product $X \circ Y = \frac{1}{2}(XY + YX)$. The trace inner product is $\langle X, Y \rangle = \text{Tr}(X \circ Y)$ and $Q(X) = \text{Tr}(X^2)$.

A.3 Sharp map and rank

The cubic norm is $N(X) = \det X$. The sharp map is

$$X^\# = \text{adj}(X), \quad X^\# \circ X = N(X) E, \quad (X^\#)^\# = N(X) X, \quad (\text{A.4})$$

with $E = \text{diag}(1, 1, 1)$. In diagonal form $X = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, $X^\# = \text{diag}(\lambda_2\lambda_3, \lambda_3\lambda_1, \lambda_1\lambda_2)$. Thus $\text{rank } X \leq 1 \iff X^\# = 0$.

A.4 Freudenthal identity

$$(X \circ Y)^\# = (N(X)Y + N(Y)X - X^\# \circ Y^\#), \quad (\text{A.5})$$

characterizing $J_3(\mathbb{O})$. Any E_6 -invariant quartic vanishing on $\{X^\# = 0\}$ is proportional to $\langle X^\#, X^\# \rangle$.

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